(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Third Semester, November 2022

Paper Code: CIC-209 **Subject: Data Structures**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1	**Attempt the following:**		
(a)	Briefly explain Abstract Data Type model with the help of a diagram.	2	CO1
(b)	Why are data structures required? List various types of data structures.	2	CO1
(c)	What is sparse matrix? Explain briefly the 3-tuple (or triplet) representation of sparse matrix with an example.	2	CO2
(d)	Differentiate between static implementation (using arrays) and dynamic implementation (using pointers) of a list data structure.	2	CO2
(e)	Define the terms depth, height, internal node, and leaf node with respect to binary trees with the help of an example.	2	CO2
2 (a)	Write algorithms for: (i) insertion of an element in a sorted singly linked list, and (ii) deletion of an element from a sorted singly linked list.	5	CO2
(b)	Consider the following Preorder and Inorder Traversals of a binary tree: Preorder : A B D E C F Inorder : D B E A F C You are required to construct the binary tree based on above given traversals. Show all the steps to construct the tree.	5	CO2
3 (a)	What is binary search tree? Write the algorithms for Preorder, Inorder and Postorder Traversals of the tree.	5	CO2
(b)	Why circular queue preferred over simple queue? Demonstrate and explain with the help of a suitable example.	5	CO2
4 (a)	Convert the following infix expression into postfix expression using stack: A + B * C – D / (E * F) ^ G	5	CO2
(b)	Construct the binary search tree using the following data elements: 35, 70, 50, 10, 48, 80, 55, 11, 85 Show binary search tree after insertion of each data element.	5	CO2

UNIT-III

Q6. a) What information is contained by the Symbol Table? Explain the contents and its capabilities of symbol table. **(7)**

b) What is error? Explain sources of errors and explain various types of errors at each phase of compiler. **(5.5)**

Q7. a) Describe various data structures of symbol tables in detail. **(5.5)**

b) What are the different storage allocation strategies in the runtime environment of the compiler? Explain. **(7)**

UNIT-IV

Q8. a) What is code generation? Explain various issues of Code Generation. **(6.5)**

b) Briefly explaina) code generation from DAGs b) Value number and algebraic laws **(6)**

Q9. a) How Peep Hole Optimization is useful in Code Optimization& Code Generation phase. Explain briefly. **(5.5)**

b) What are basic blocks and Flow Graphs? Explain with the help of examples. **(7)**

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